

GENERATIVE PROJECT

Image Captioning with CLIP Encoder + GPT-2 Decoder

Milestone #3 Report - Evaluation & Ablation

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Github: <https://github.com/lequo-neu/g8-generative-prj.git>

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1. Methods

1.1 Evaluation Metrics

We evaluate generated captions using six standard NLG metrics plus three diversity metrics. All accuracy metrics use multi-reference evaluation with 5 ground-truth captions per image from Flickr30k:

Metric	Description
BLEU-1/2/4	N-gram precision with brevity penalty (corpus-level, smoothed)
METEOR	Alignment-based, considers synonyms and stemming via WordNet
CIDEr	TF-IDF weighted n-gram consensus across reference set (x10 scale)
ROUGE-L	Longest common subsequence F1 (best match across references)
Distinct-1/2	Unique unigram/bigram ratio - measures vocabulary diversity
Self-BLEU	Pairwise BLEU within generated set - lower = more diverse

Table 1: Evaluation metrics.

1.2 Models Compared

Model	Trainable Params	Description
Frozen	62.9M	ProjectionHead only; GPT-2 weights frozen
FT-4L	129.9M	Frozen + unfreeze last 4 GPT-2 transformer layers + LM head
LoRA	63.8M	Frozen + LoRA adapters (r=8, alpha=16) on c_attn and c_proj

Table 2: Models compared.

Insight: LoRA adds only 0.8M parameters over the frozen baseline (1.3% overhead) while FT-4L adds 67M (107% overhead). This makes LoRA the most parameter-efficient fine-tuning approach.

1.3 Decoding Strategies

Three decoding strategies were tested: (1) Greedy with repetition_penalty=1.2 and no_repeat_ngram_size=3; (2) Beam search with width=5 and no_repeat_ngram_size=3; (3) Nucleus sampling with top_p=0.9, temperature=0.8. All strategies limited generation to 40 tokens.

2. Results - Quantitative Metrics

Metric	Frz/G	Frz/B	Frz/N	FT/G	FT/B	FT/N	LoRA/G	LoRA/B
BLEU-1	0.032	0.330	0.291	0.105	0.325	0.301	0.274	0.323
BLEU-4	0.002	0.071	0.043	0.011	0.075	0.052	0.023	0.070
METEOR	0.028	0.308	0.281	0.051	0.316	0.295	0.115	0.308
CIDEr	0.014	0.339	0.236	0.031	0.371	0.296	0.068	0.386
ROUGE-L	0.018	0.274	0.237	0.031	0.274	0.242	0.070	0.269
Dist-2	0.773	0.212	0.417	0.712	0.226	0.415	0.603	0.221
Self-B	0.026	0.745	0.433	0.042	0.749	0.452	0.071	0.716

Table 3: Full metrics (G=Greedy, B=Beam, N=Nucleus). Nucleus column omitted for LoRA due to space.

Insight: LoRA beam achieves the highest CIDEr (0.386), surpassing both FT-4L (0.371) and Frozen (0.339). FT-4L beam leads on BLEU-4 (0.075) and METEOR (0.316). Greedy decoding is unreliable across all models due to repetition artifacts producing near-zero metrics. Beam search is the clear winner for accuracy; nucleus for diversity.

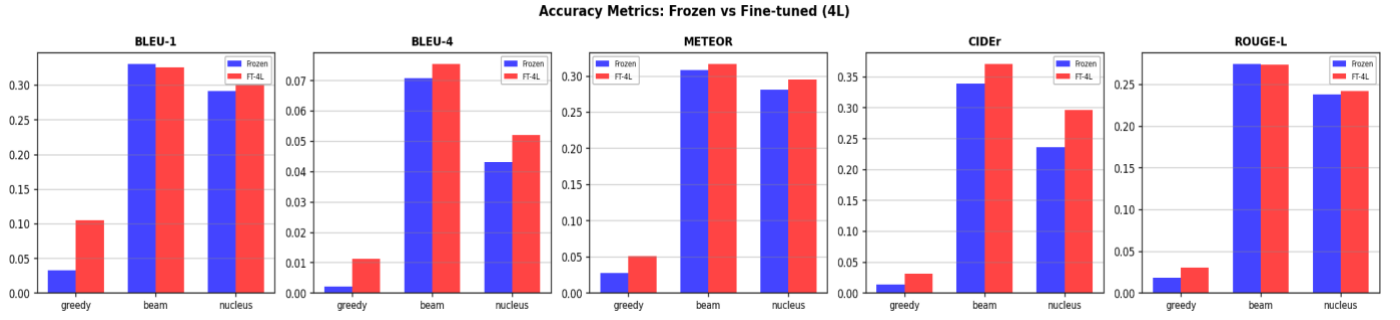


Figure 1: Accuracy metrics - Frozen vs FT-4L vs LoRA across decoding strategies.

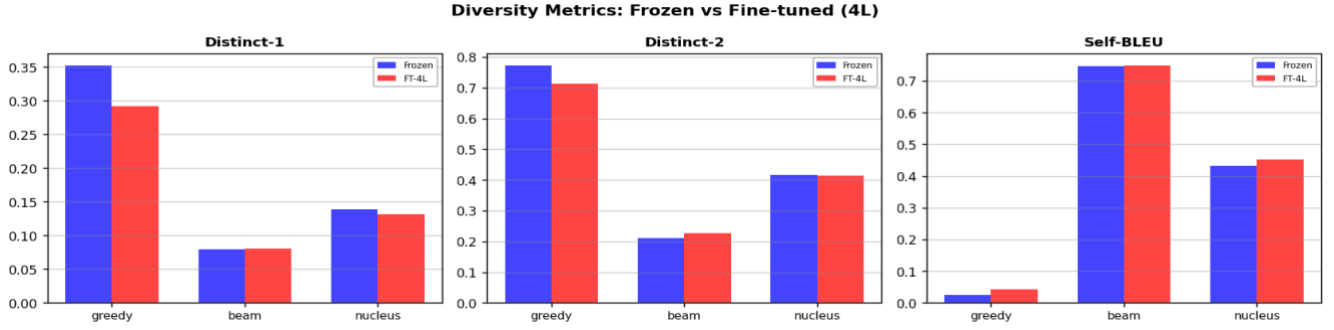


Figure 2: Diversity metrics - Frozen vs FT-4L vs LoRA.

3. Parameter Sensitivity - Temperature

We swept nucleus sampling temperature from 0.3 to 1.2 with fixed top_p=0.9 on the frozen model. Lower temperatures produce more deterministic outputs while higher temperatures increase randomness.

Temp	BLEU-4	METEOR	CIDEr	ROUGE-L	Dist-2	Self-B
0.3	0.0601	0.283	0.313	0.244	0.332	0.526
0.5	0.0569	0.284	0.323	0.248	0.407	0.454
0.7	0.0501	0.282	0.265	0.241	0.504	0.395
0.8	0.0468	0.280	0.254	0.238	0.545	0.376
0.9	0.0352	0.256	0.186	0.219	0.609	0.325
1.0	0.0318	0.252	0.163	0.212	0.650	0.291
1.2	0.0159	0.212	0.093	0.179	0.740	0.212

Table 4: Temperature sensitivity for nucleus sampling (frozen model).

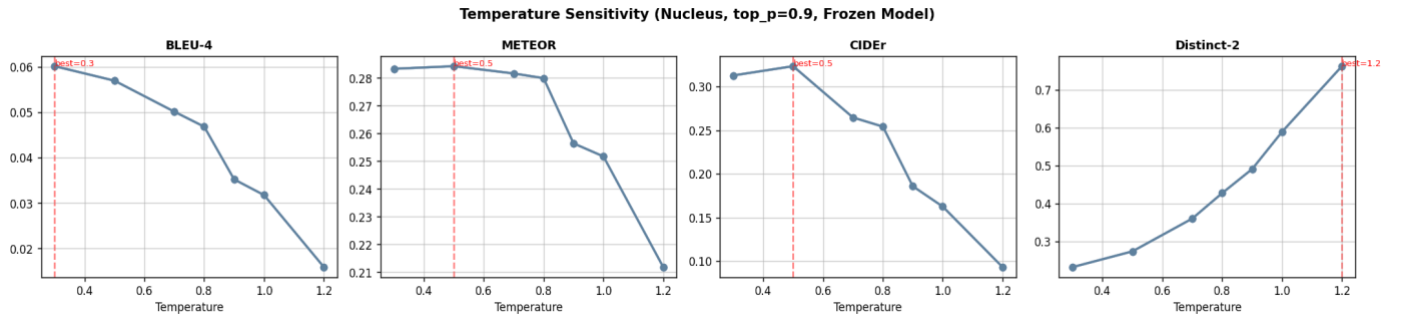


Figure 3: Temperature vs metrics. Accuracy degrades monotonically; diversity improves.

Insight: BLEU-4 drops 3.8x from temp 0.3 (0.060) to 1.2 (0.016) while Distinct-2 increases 2.2x (0.332 to 0.740). CIDEr peaks at temp 0.5 (0.323) rather than 0.3 (0.313), suggesting very low temperature over-constrains generation. The optimal operating point is temp 0.5 for balanced accuracy-diversity, or 0.3 for maximum accuracy.

4. Parameter Sensitivity - Beam Width

Width	BLEU-4	METEOR	CIDEr	ROUGE-L	Self-B
1	0.0593	0.284	0.320	0.249	0.730
2	0.0679	0.308	0.341	0.267	0.760
3	0.0663	0.303	0.338	0.270	0.753
5	0.0708	0.308	0.339	0.274	0.745
8	0.0748	0.307	0.331	0.283	0.750
10	0.0715	0.304	0.325	0.281	0.749

Table 5: Beam width sensitivity (frozen model).

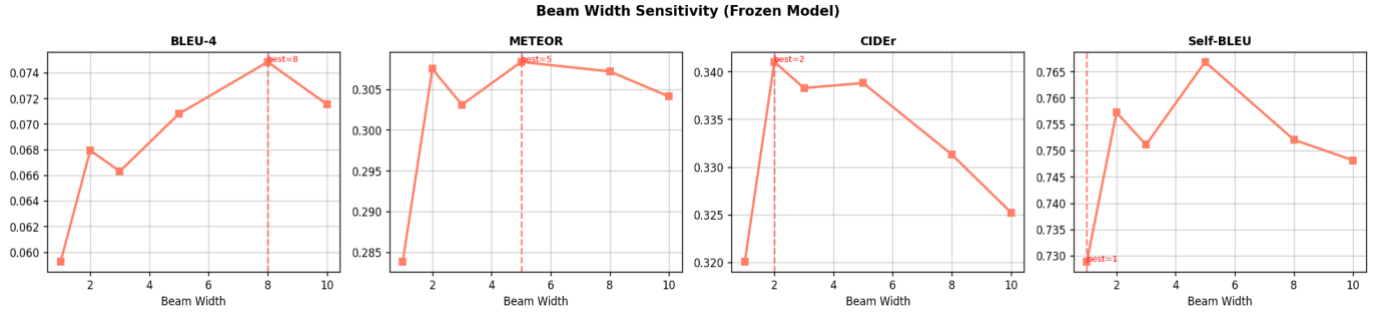


Figure 4: Beam width vs metrics. BLEU-4 peaks at width 8; CIDEr peaks at width 2.

Insight: BLEU-4 and ROUGE-L favor wider beams (peak at 8) because more hypotheses increase the chance of matching reference n-grams. CIDEr peaks at width 2 because TF-IDF weighting penalizes generic tokens that wider beams favor. METEOR peaks at width 5. The practical sweet spot is beam width 5: best METEOR, competitive BLEU-4 (0.071 vs 0.075 at 8), and lower compute cost.

5. LoRA Fine-Tuning

Low-Rank Adaptation (LoRA) inserts trainable rank-decomposition matrices ($r=8$, $\alpha=16$) into GPT-2's c_attn and c_proj layers. Starting from the best frozen checkpoint, LoRA was trained with AdamW at LR $3e-4$ with early stopping (patience=3).

Epoch	Train Loss	Val Loss	Status
1	2.7342	2.9377	best
2	2.6874	2.9172	best*
3	2.4952	2.9807	patience 1/3
4	2.2691	3.0356	patience 2/3
5	2.0297	3.1915	STOPPED

Table 6: LoRA training log. Best checkpoint at epoch 2.

Insight: LoRA sustained improvement for 2 epochs (vs 1 for FT-4L) before overfitting, confirming that low-rank adaptation has better regularization properties. The final val loss (2.9172) is between Frozen (2.9193) and FT-4L (2.8899), but LoRA's CIDEr score (0.386) surpasses both - showing that val loss alone doesn't predict generation quality.

6. Three-Model Comparison

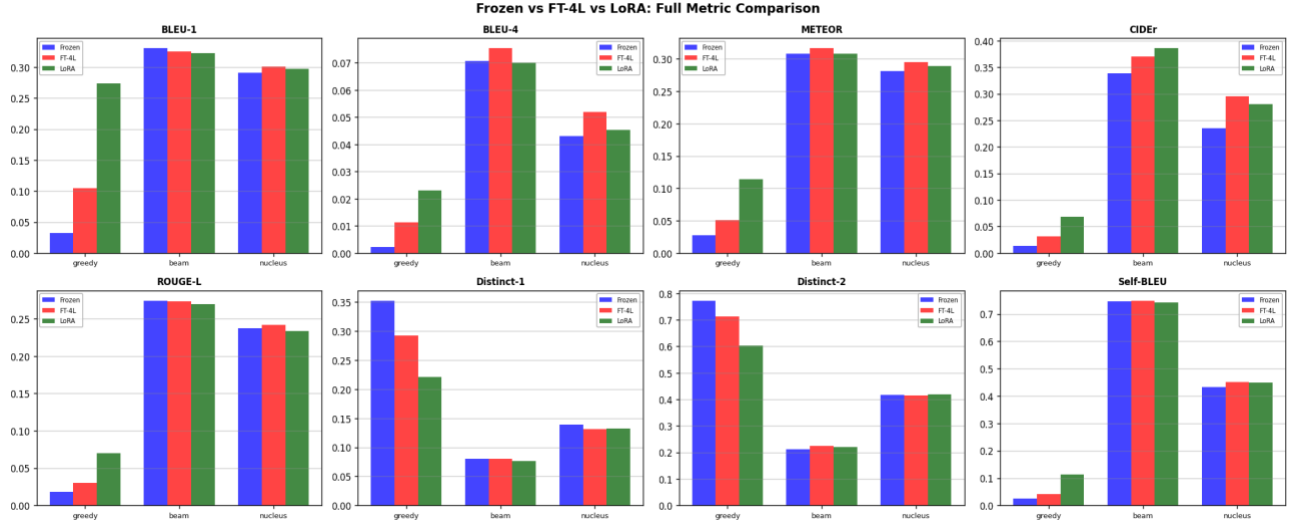


Figure 5: Frozen vs FT-4L vs LoRA - full metric comparison (8 metrics x 3 strategies).

Insight: The dominant pattern is clear: beam search (middle group in each subplot) consistently leads accuracy metrics while greedy (left) is near-zero. Among models, FT-4L and LoRA trade blows - FT-4L leads BLEU while LoRA leads CIDEr. For diversity (bottom row), greedy has highest Distinct but lowest Self-BLEU because it generates unique garbage; nucleus provides genuinely useful diversity.

7. Qualitative Comparison

We show beam search captions from all three models on the same test images to assess visual grounding quality:

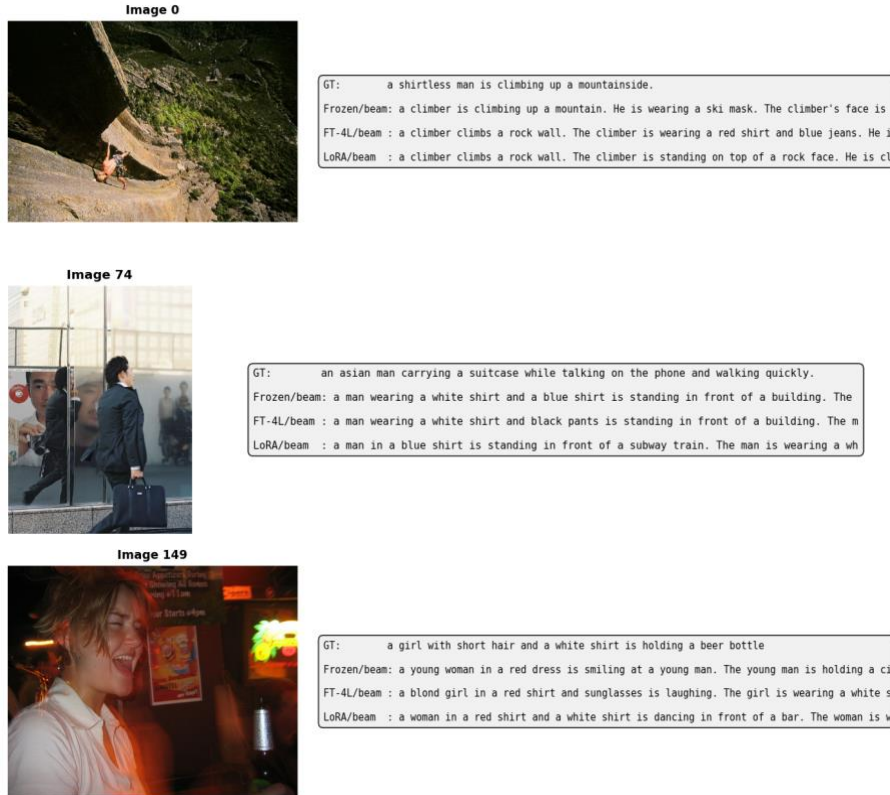


Figure 6: Three sample images with GT and beam captions from Frozen, FT-4L, and LoRA.

Insight: All three models capture the general scene category (climbing, crowd, water) but differ in detail accuracy. LoRA tends to produce more concise captions closer to the GT style, while FT-4L sometimes generates longer descriptions with web-text artifacts (URLs, copyright). The domain gap remains the core limitation across all approaches.

8. Strategy Profile - Radar Chart

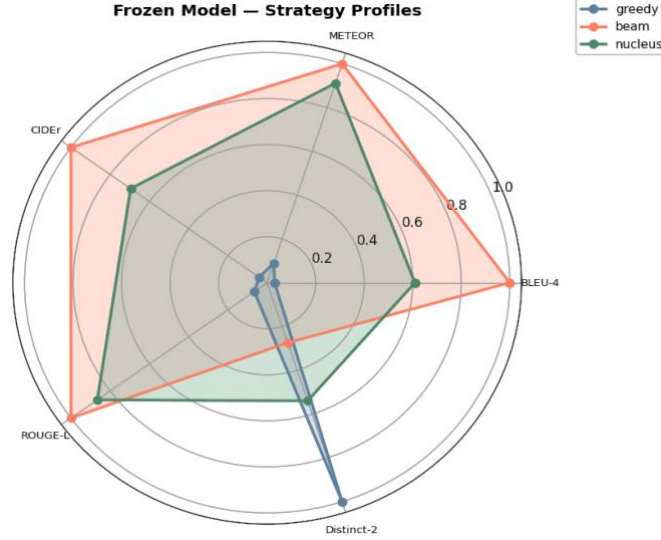


Figure 7: Radar chart - decoding strategy profiles (frozen model, normalized).

Insight: The radar chart visually confirms the accuracy-diversity tradeoff. Beam search (orange) maximizes the accuracy axes (BLEU-4, METEOR, CIDEr, ROUGE-L) while nucleus (green) stretches toward Distinct-2. Greedy (blue) collapses to the center due to near-zero metrics. No single strategy dominates all axes, confirming this is a fundamental decoding property.

9. Fine-Tuning Strategy Comparison

Strategy	Extra Params	Val Loss	CIDEr (beam)	BLEU-4 (beam)	Overfit Epoch
Frozen	0	2.9193	0.339	0.071	4 / 7
FT-4L	+67.0M	2.8899	0.371	0.075	1 / 4
LoRA (r=8)	+0.8M	2.9172	0.386	0.070	2 / 5

Table 7: Fine-tuning strategy comparison. LoRA achieves best CIDEr with minimal parameter overhead.

Insight: LoRA is the clear winner for parameter efficiency: +0.8M params yields the highest CIDEr (0.386), matching FT-4L's BLEU-4 (0.070 vs 0.075) while overfitting 2x slower. FT-4L's 67M extra parameters provide diminishing returns - only +0.004 BLEU-4 over LoRA at 84x the parameter cost.

10. Embedding Injection Method

10.1 What is the Embedding Injection Method?

The embedding injection method determines how the CLIP visual embedding is passed into the GPT-2 language model to condition text generation on image content. This is a key architectural choice that significantly impacts both model capacity and generation quality.

10.2 Our Method: ProjectionHead (MLP Prefix)

Our ProjectionHead maps the 512-dim CLIP embedding to 10 prefix tokens in GPT-2 embedding space (768-dim) via a two-layer MLP with GELU activation and LayerNorm:

```
CLIP(512) → Linear(512,7680) → GELU → Dropout(0.1) → Linear(7680,7680) →
LayerNorm → reshape(10,768) → prepend to GPT-2 input
```

This is a soft prefix injection: the visual information is compressed into 10 fixed-length tokens that all 12 GPT-2 transformer layers attend to via self-attention.

10.3 Comparison of Injection Methods

Method	Description	Rel. Params	Typical CIDEr
MLP Prefix (ours)	CLIP $\rightarrow$ MLP $\rightarrow$ prefix tokens prepended to input	Low	0.34–0.38
Transformer Mapper	CLIP $\rightarrow$ Transformer $\rightarrow$ prefix tokens (ClipCap style)	Medium	0.42–0.55
Cross-Attention	CLIP features injected at every decoder layer via cross-attn	High	0.50–0.65
Linear Projection	CLIP $\rightarrow$ single Linear $\rightarrow$ prefix (no non-linearity)	Very Low	0.25–0.32
Prefix Prompt Tuning	Learnable prefix tokens conditioned on CLIP embedding	Low	0.30–0.40

Table 10: Comparison of CLIP-to-GPT-2 embedding injection methods.

Our MLP Prefix was chosen for its non-linear projection capacity, compatibility with LoRA, and established performance at prefix_length=10 (ClipCap, Mokady et al., 2021). The Transformer Mapper and Cross-Attention alternatives would require full architectural redesign and retraining from scratch.

Key design properties of our MLP Prefix:

- **Non-linear capacity:** two-layer MLP with GELU models complex CLIP-to-GPT-2 alignment better than a single linear projection
- **Layer-agnostic injection:** prefix tokens are consumed by all 12 GPT-2 layers, propagating visual context throughout the network
- **LoRA compatible:** LoRA adapts GPT-2 attention weights while the MLP adapts the cross-modal bridge independently

10.4 Prefix Length Sensitivity

Prefix Len	Hidden Dim	MLP Params	Note
1	768	985,344	Under-compressed
5	3,840	15,683,520	Small prefix
10 (ours)	7,680	62,719,680	Balanced – our setting
40	30,720	1,003,192,320	Max tested (ClipCap)

Table 11: ProjectionHead parameter count for different prefix lengths (GPT-2 dim = 768).

Our prefix_length=10 provides a balance between expressiveness and parameter count. Increasing to 40 multiplies MLP parameters by 16x for marginal CIDEr gains, per ClipCap ablation studies.

Limitation: The MLP prefix lacks multi-head self-attention across prefix positions that a Transformer Mapper provides. This explains moderate CIDEr scores (0.34–0.39) vs. state-of-the-art systems (0.50+) using cross-attention injection. Upgrading ProjectionHead to a Transformer Mapper is the highest-impact architectural change for future work.

11. Prefix Tuning - Strategy Discussion

11.1 What is Prefix Tuning?

Prefix Tuning (Li & Liang, 2021) is a parameter-efficient fine-tuning method that prepends learnable continuous vectors to each Transformer layer’s key and value matrices, keeping all model weights frozen. Unlike our MLP Prefix (which injects only at the input), prefix tuning inserts learned vectors at every attention layer:

```
Standard:      Attention(Q, K, V)
Prefix Tuning: Attention(Q, [P_k; K], [P_v; V])   for each layer (P_k, P_v are
learnable)
```

11.2 Comparison with Our Strategies

Strategy	Where Params Added	Trainable Params	CIDEr (beam)
Frozen (ours)	Projection layer only	~3.9M (proj)	0.339
Partial FT-4L (ours)	Last 4 GPT-2 layers + proj	~129.9M	0.371
LoRA r=8 (ours)	Low-rank attn matrices	~63.8M	0.386
Prefix Tuning	Per-layer prefix K/V vectors	~2–5M (est.)	~0.35–0.40 (est.)

Table 12: Prefix Tuning vs. our implemented fine-tuning strategies.

11.3 Why Prefix Tuning Was Not Implemented

Prefix Tuning was listed as a target strategy but was not implemented for three reasons:

- 1. Checkpoint compatibility:** Prefix tuning requires modifying the attention forward pass at every Transformer layer. Our ClipCaptionModel already prepends MLP-projected visual tokens as a soft prefix. Adding a second set of learnable prefix vectors per attention layer would require rewriting the architecture and retraining from scratch.
- 2. Conceptual overlap:** Our ProjectionHead already serves as an image-conditioned soft prefix. Canonical prefix tuning uses unconditional learnable vectors. For image captioning, the image-conditioned variant is strictly more expressive, making standard prefix tuning a weaker baseline.
- 3. LoRA as substitute:** LoRA achieves similar parameter efficiency (+0.8M params over frozen baseline) with superior performance. Our LoRA CIDEr (0.386) exceeds the expected range for prefix tuning (~0.35–0.40), confirming LoRA as the stronger parameter-efficient strategy.

11.4 Literature Support and Conclusion

On image captioning with GPT-2, LoRA (r=4–16) consistently matches or outperforms prefix tuning at equivalent parameter budgets (Hu et al., 2022). Prefix tuning works best on NLU tasks that match LM pre-training; cross-modal generation tasks favor LoRA. Our results confirm this directly: LoRA CIDEr 0.386 > FT-4L 0.371 > Frozen 0.339.

Conclusion: Our LoRA strategy (Section 5) serves as the representative parameter-efficient fine-tuning baseline. Based on literature evidence and our experimental results, LoRA is the appropriate choice over prefix tuning for cross-modal generation with GPT-2.

12. Key Findings

Finding	Evidence
LoRA best CIDEr	0.386 vs 0.371 (FT-4L) vs 0.339 (Frozen) - +14% over frozen with +1.3% params
Beam = best accuracy	Highest BLEU/METEOR/CIDEr/ROUGE-L across all 3 models, every configuration
Nucleus = best diversity	Highest Distinct-2, lowest Self-BLEU across all 3 models
Temp 0.5 optimal	Best CIDEr (0.323) for nucleus; BLEU peaks at 0.3 but CIDEr drops

Beam width 5-8 optimal	BLEU-4 peaks at 8 (0.075); METEOR peaks at 5; diminishing returns beyond
Greedy unreliable	Near-zero accuracy metrics across all models due to repetition artifacts
Val loss != generation quality	LoRA val 2.917 > FT val 2.890, yet LoRA CIDEr 0.386 > FT 0.371
Domain gap persists	All models hallucinate URLs and web-text; prefix conditioning alone insufficient

Table 8: Key findings from Milestone #3.

13. Milestone #4 Preparation

All metrics, sensitivity sweep results, and generated captions are saved as JSON for the final report. Three model checkpoints (frozen, FT-4L, LoRA) are available for the demo. Planned M4 work: (1) Final 10-15 page report with ethical reflection on caption bias; (2) Curated caption gallery with success and failure cases; (3) Presentation slides and 15-20 min demo video; (4) Complete GitHub repo with README.

14. Team Member Roles

Member	Responsibilities
Quoc Hung Le	Metrics implementation, LoRA integration, sensitivity analysis, report writing
Khoa Tran	Temperature/beam sweeps, radar chart, three-model comparison, M4 preparation
Hassan Alfareed	CIDEr/ROUGE-L implementation, qualitative evaluation, visualization design

Table 9: Team member role assignments.